EE 344 Project Proposal

Robot Execution Failure Detection via Unsupervised Learning

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1) Application Area

This project focuses on robotic manufacturing and automation, specifically detecting and characterizing execution failures in industrial robot arms during assembly tasks. Robots performing pick-and-place operations can fail due to collisions, slippage, misalignment, or unexpected contact forces. Rapid identification of failure types is critical for reducing downtime and enabling adaptive recovery. Rather than relying on brittle hand-crafted rules, we aim to use data-driven methods to discover meaningful failure groupings directly from force/torque sensor data.

2) Dataset Selection

- **Dataset:** Robot Execution Failures Data Set
- **Source:** UCI Machine Learning Repository https://archive.ics.uci.edu/dataset/138/robot+execution+failures
- **Authors:** L. Seabra Lopes and L.M. Camarinha-Matos, Universidade Nova de Lisboa (1998)

The dataset contains force and torque measurements recorded from a robot arm after failure detection during assembly operations. It includes **5 sub-datasets** (LP1–LP5), each corresponding to a different phase of the assembly task:

Sub-dataset	Task Phase	Samples	Classes
LP1	Approach to grasp position	88	4
LP2	Transfer of a part	47	5
LP3	Position after transfer failure	47	4

Sub-dataset	Task Phase	Samples	Classes
LP4	Approach to ungrasp position	117	3
LP5	Motion with part	164	5

Each sample has **90 features**: 15 time steps × 6 sensor channels (F_x, F_y, F_z, T_x, T_y, T_z). The observation window per failure instance is 315 ms. All features are integer-valued. We plan to primarily use **LP4 and LP5** (the two largest sub-datasets) and explore combining all five for a more comprehensive analysis.

Limitations: The dataset is relatively small (463 total samples across all sub-datasets), which may limit model complexity. There are no missing values, but class distributions are imbalanced within each sub-dataset. The data was collected under controlled laboratory conditions, which may limit generalizability to other robotic platforms.

3) Data Type Characterization

Category	Characterization
Structure	Structured (tabular, fixed-length feature vectors)
Labels	Labeled (failure-type labels available; used only for validation, not training)
Data types	Quantitative (all integer-valued force/torque measurements)
Temporal	Time-dependent (each sample is a 15-step time series of 6 sensor channels)

4) Data Science / Machine Learning Problem Definition

Task type: Unsupervised clustering and anomaly detection. The goal is to discover meaningful groupings of robot execution failures from force/torque sensor data without using class labels during training. Labels are reserved for evaluation only.

Inputs (X): 90-dimensional feature vectors (15 time steps × 6 channels: F_x, F_y, F_z, T_x, T_y, T_z), plus engineered features (per-channel statistics, FFT-based frequency features, temporal gradients).

Unsupervised objective: Identify natural failure groupings. Secondary task: anomaly detection — train on the most common failure type and detect rare

failures as outliers.

What success looks like: Clusters that align well with ground-truth failure labels (measured by Adjusted Rand Index and Normalized Mutual Information), demonstrating that unsupervised methods can reliably separate failure modes without labeled training data.

5) Why Data Science / Machine Learning?

Simple rule-based approaches (e.g., thresholding individual sensor channels) fail here because failure modes produce complex, overlapping patterns across multiple channels simultaneously. The discriminative information lies in the joint *temporal patterns* across all six channels. Unsupervised ML offers key advantages: (1) it discovers structure without expensive manual labeling; (2) it can generalize to unanticipated failure types; and (3) dimensionality reduction provides interpretable visualizations that help engineers understand the failure landscape.

6) Planned Approaches

(a) Preprocessing / Feature Engineering

- **Scaling:** StandardScaler and MinMaxScaler — compare impact on clustering.
- **Statistical features:** Per-channel summary statistics (mean, std, max, min, skewness, kurtosis).
- **Frequency-domain features:** FFT per channel; extract dominant frequency magnitudes and spectral energy.
- **Dimensionality reduction:** PCA (retain 95% variance) and t-SNE/UMAP for visualization.
- **Outlier handling:** Inspect and document extreme values; compare results with/without outlier removal.

(b) Learning Methods / Models

- **K-Means Clustering:** Baseline partitioning method; use elbow method and silhouette analysis to select k.

- **Gaussian Mixture Models (GMM):** Probabilistic clustering with soft assignments; compare covariance structures (full, diagonal, tied) and use BIC/AIC for model selection.
- **Autoencoder + Clustering:** Shallow autoencoder for nonlinear dimensionality reduction, then K-Means or DBSCAN in the latent space.
- **(Stretch) Isolation Forest / One-Class SVM:** Frame as anomaly detection — train on majority class, detect minority failures as outliers.

7) Evaluation Plan

Internal metrics (unsupervised): Silhouette Score, Davies-Bouldin Index, Calinski-Harabasz Index — assess cluster quality without labels.

External metrics (label-based validation): Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and cluster purity — measure alignment with known failure types.

For anomaly detection: Precision, Recall, F1-score, ROC-AUC.

Evaluation strategy: We will use repeated random sub-sampling (5-fold) on the combined LP4+LP5 dataset and report results across multiple hyperparameter settings and random seeds. The **primary metric** is ARI (clustering) and Silhouette Score (internal validation); supporting metrics include NMI and cluster purity.

8) Team Plan

Task	Owner	Timeline
Data loading, EDA, visualization	Ye Cao	Week 6–7
Preprocessing pipelines & feature engineering	Ye Cao	Week 6–7
K-Means & GMM implementation + tuning	Ye Cao	Week 7–8
Autoencoder + latent-space clustering	Ye Cao	Week 7–8
Anomaly detection (stretch goal)	Ye Cao	Week 8–9
Evaluation, comparison, & figures	Ye Cao	Week 9
Final report & presentation	Ye Cao	Week 10